SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

EAS109

Examination - January Semester 2023

Thermo-Fluid Mechanics

Thursday, 18 May 2023 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SIX (6)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Full marks: 100)

Question 1

An airplane is cruising an altitude of 10,000 m above sea level. The flow of air over the wing of the airplane, as shown in Figure Q1 below, is at 300 km/hr. The air approaching the wing at 65 kPa and -15°C at point 1. At point 2 in the flow, the pressure is 60 kPa.

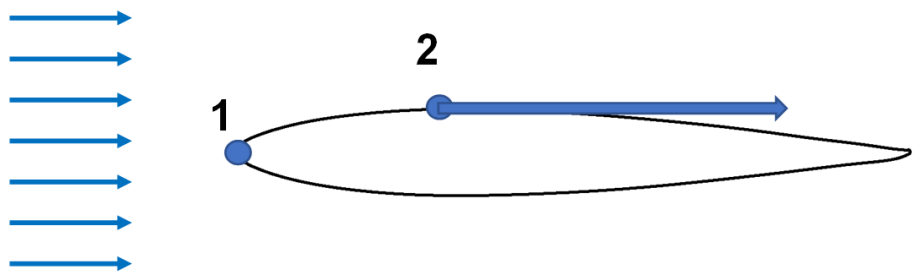

Figure Q1

Question 1a

From Bernoulli's Equation, recognise the local flow speed along the streamline is given as:

$$V_{local} = \sqrt{\frac{2(P_{stag} - P)}{\rho}}$$

where

P_{stag} - stagnation pressure,

P - static pressure,

ρ - density of the fluid

V_{local} - local flow speed of the fluid

Take air as an ideal gas (PV = mRT) with the constant value of R = 0.287 kJ/kgK. [Note that here V denotes volume]

(5 marks)

Question 1b

Apply the Bernoulli Equation and the Ideal Gas Law to evaluate the following stated in Q1b(i) and Q1b(iv):

Question 1b(i)

Air density at the cruising altitude in kg/m^3.

(2 marks)

Question 1b(ii)

Speed of the air relative to the wing at point 2 in m/s.

(5 marks)

Question 1b(iii)

Stagnation pressure at point 1 in kPa.

(5 marks)

Question 1b(iv)

Absolute air speed in m/s.

(5 marks)

Question 1c

Identify and explain **THREE (3)** assumptions you made in your results.

(3 marks)

Question 2

A horizontal jet of water at a velocity of 60 m/s from a nozzle enters a stationary curved vane tangentially at point ① that deflect the water jet down at an angle of 90°, as shown in Figure Q2 below. The water leaves the vane at point ② with a velocity of 45 m/s. If the diameter of the nozzle exit is 10 mm:

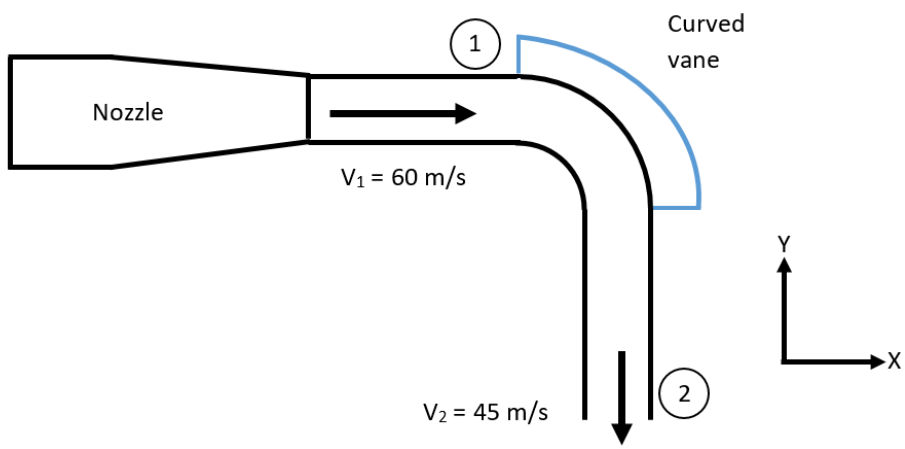

Figure Q2

Question 2a

Applying Newton's 2nd law of motion, recognise that the expression of force on gas as:

$$F = \dot{m}(V_2 - V_1)$$

(5 marks)

Question 2b

Evaluate the mass flow rate of water discharge from the nozzle (in kg/s).

(2 marks)

Question 2c

Based on the X and Y direction shown, interpret the force (in N) in the X and Y directions acting on the water. Indicate whether the forces are positive or negative.

(8 marks)

Question 2d

Evaluate the magnitude and direction of the resultant force acting on the vane. Sketch the vector diagram, clearly labelling the resultant forces of the water and on the vane.

(10 marks)

Question 3

A private jet with a wing area of 16.4 m^2 weights approximately 1.5 tonnes. At take-off, the lift coefficient (C_L) and drag coefficient (C_D) are 0.9 and 0.05 respectively.

Question 3a

Evaluate and interpret the take-off speed in km/h, Lift and Drag forces during the take-off.

(12 marks)

Question 3b

Once in cruising at a speed of 300 km/h, interpret the required power in kW if the cruising drag coefficient (C_D) is 0.035.

(8 marks)

Question 3c

Identify and describe **THREE (3)** assumptions with regards to the flow and **ONE (1)** error towards the calculated drag force.

(5 marks)

Question 4

A gas turbine engine operating on a Brayton Cycle with air as the working fluid has a pressure ratio of 8 and maximum cycle temperature of 800°C. The Isentropic efficiencies of the compressor and turbine are 0.83 and 0.87 respectively. Air enters the turbine at 20°C and pressure of 1 bar.

[Take Cp_{air} = 1.005 kJ/kgK, Cp_{gas} = 1.15 kJ/kgK; γ_{air} = 1.4, γ_{gas} = 1.333]

Question 4a

From the Ideal Gas (PV = mRT) and Polytropic Process (PV^{γ} = constant) equations, identify by derivation

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{(\gamma-1)}{\gamma}}$$

(5 marks)

Question 4b

Describe by sketching the Temperature-Entropy of the Brayton Cycle.

(5 marks)

Question 4c

Compute and interpret the mass flowrate of air required to produce an output power of 25 MW.

(10 marks)

Question 4d

Evaluate the gas turbine's cycle thermal efficiency and its work ratio.

(5 marks)

----- END OF PAPER -----